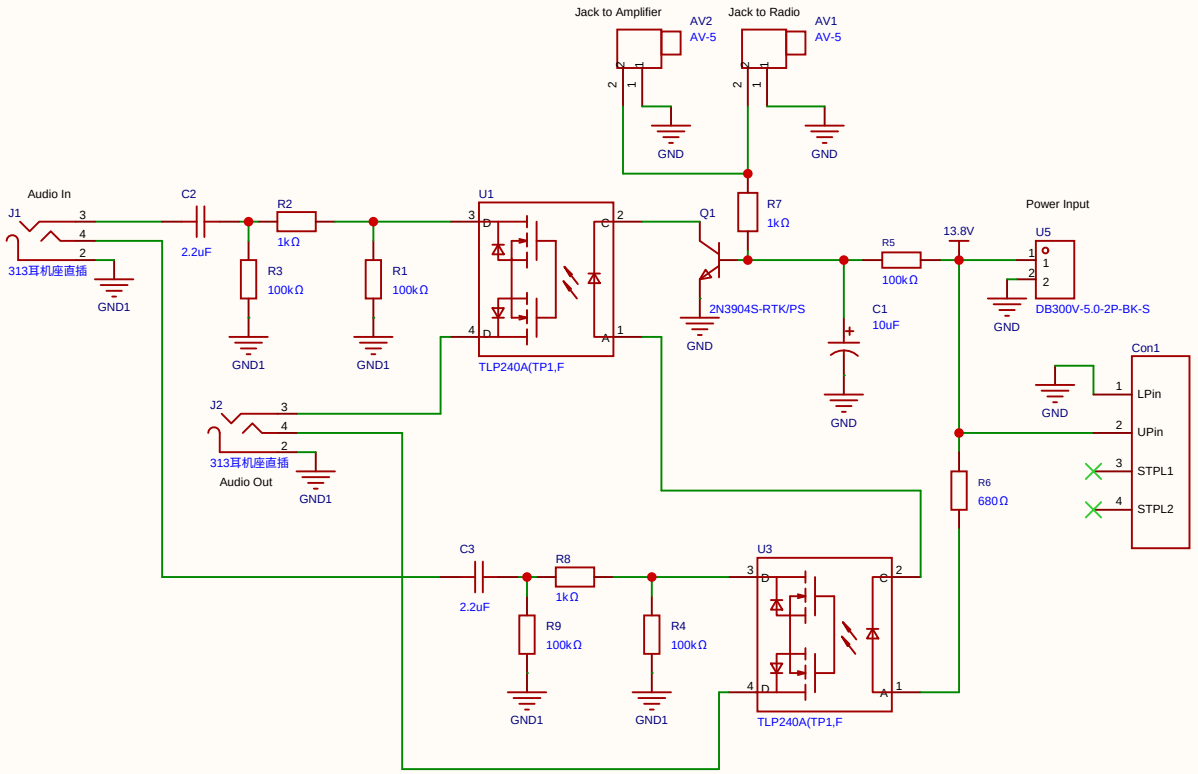


This circuit is my take on a solid state version of the relay version found at:
https://www.sdrutah.org/info/ptt_mute.html .
The site doesn't list who designed the relay version.

K0WLY

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CIRCUIT OPERATION & NOTES

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1. APPLICATION: Dual-channel (Stereo) line-level audio mute for RF transmit keying. Provides soft fade-in/fade-out to eliminate switching pops/clicks.

2. INPUT/OUTPUT ISOLATION: Audio signals are AC-coupled through 10µF caps \(C_IN\). Dual PhotoMOS optocouplers (AQY212EHA/TLP240A) provide solid-state audio switching with complete optical isolation between control and audio paths. 100KΩ bleed resistors maintain 0V DC reference across switch terminals. If the hFE < 120 on your transistor, then change the 100k resistor to 47KΩ.

3. CONTROL LOGIC & RC FADE:
STANDBY (Radio Unkeyed / RCA Open): R_PULLUP (100kΩ) holds 2N3904 base HIGH (above ~0.7V), driving ~10mA through series PhotoMOS LEDs. Audio passes.
TRANSMIT (Radio Keyed / RCA Grounded): Radio grounds the RCA center pin, discharging C_FADE \(10µF\) through R_LIMIT \(1kΩ). This ramps the NPN base voltage down over ~30 – 50ms, softly fading the PhotoMOS junction from low resistance to high resistance to mute audio without transient clicks.

4. AMP PASS-THROUGH: RCA Jack 1 (Radio SEND) and RCA Jack 2 (Amp Keying) are wired directly in parallel for low-side pass-through keying to external gear.

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Schematic	Schematic1	Create at	2026-08-27
		Update at	2026-10-08
Board	Board1	Page	P1
Drawn	Cowley (K0WLY)	SDR_Audio_Switch	
Reviewed			
		Version / Size / Page 1 Total 1	
EasyEDA		V1.0 / A4	EasyEDA.com